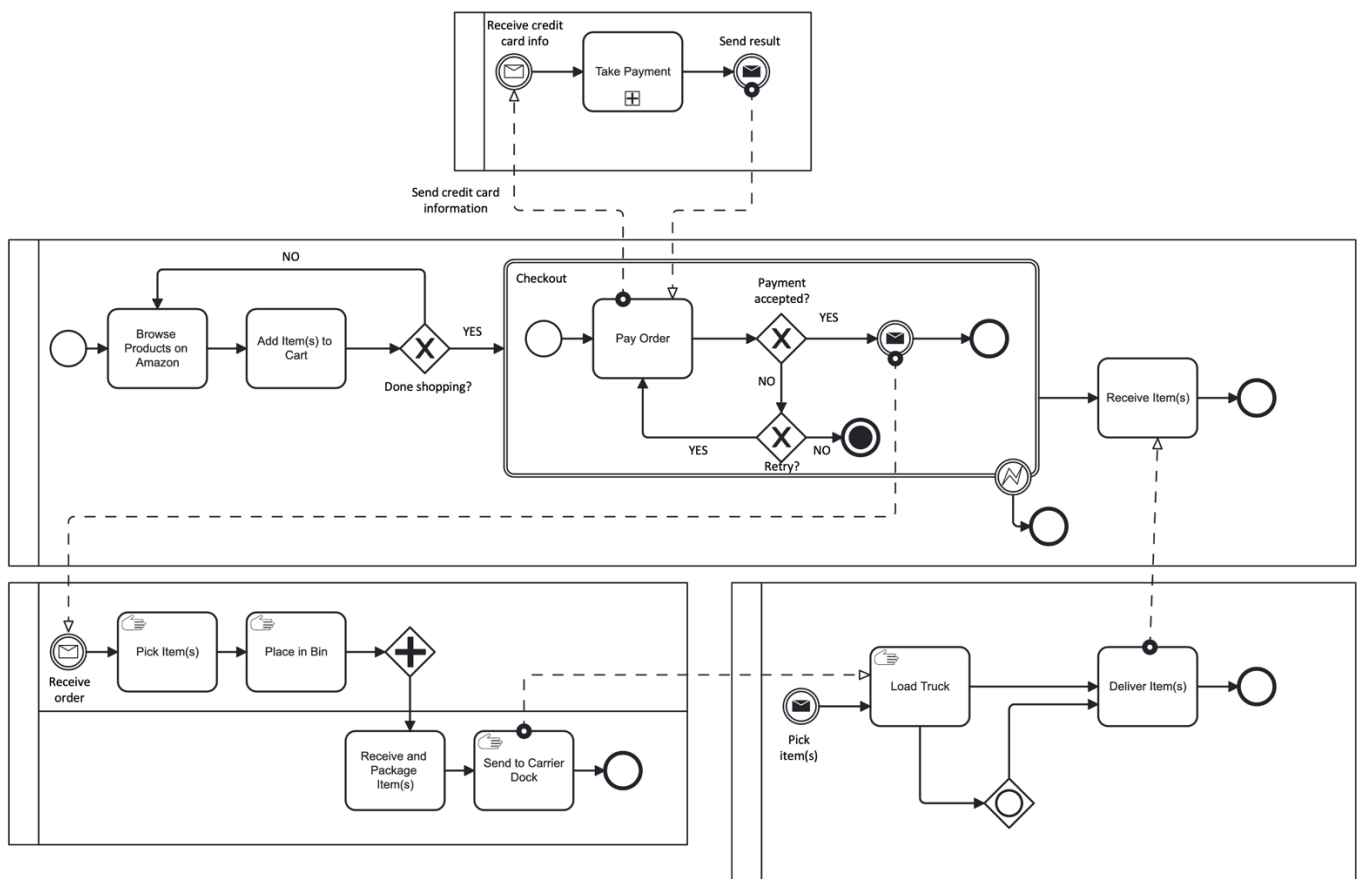


The Amazon case

The online shopping on Amazon begins with the customer browsing Amazon's product offerings, selecting items of interest, and adding them to the shopping cart. Once the desired items are in the cart, the customer proceeds to the checkout stage, where they enter their payment information. This payment data is securely transmitted to the credit card company for authorization. In that stage, the credit card company processes the details of the order against the credit limits of the card. If the payment is approved, Amazon's system confirms the order and forwards it to the order fulfillment department. However, if the payment is declined, the customer is given the option to retry with a different payment method or correct any issues. After an order is confirmed, Amazon's fulfillment team springs into action. The first step involves the Amazon picker, who locates each item in the warehouse based on the order details. The picker then places the items into a designated bin, ready for the next stage. The bin is then passed along to the Amazon packager, who ensures that all items are properly packed and labeled for shipping. Once packaged, the items are sent to the carrier dock, where the carrier takes over. The carrier collects the package, loads it onto a delivery truck, and begins the journey to the customer's location. After a successful delivery, the customer receives their ordered items, bringing the process to a close. This streamlined flow involves multiple participants and decision points, each of which is carefully coordinated to ensure timely and accurate order fulfillment.

The human roles play a critical part in ensuring the smooth processing and fulfillment of the customer's order. Amazon prioritizes efficiency, and therefore, the BPMN modeling of this process was assigned to one of the process development teams. The task was to visualize the interactions between various roles, so that the organization can identify areas that efficiency can be improved. The team provided the following BPMN model.



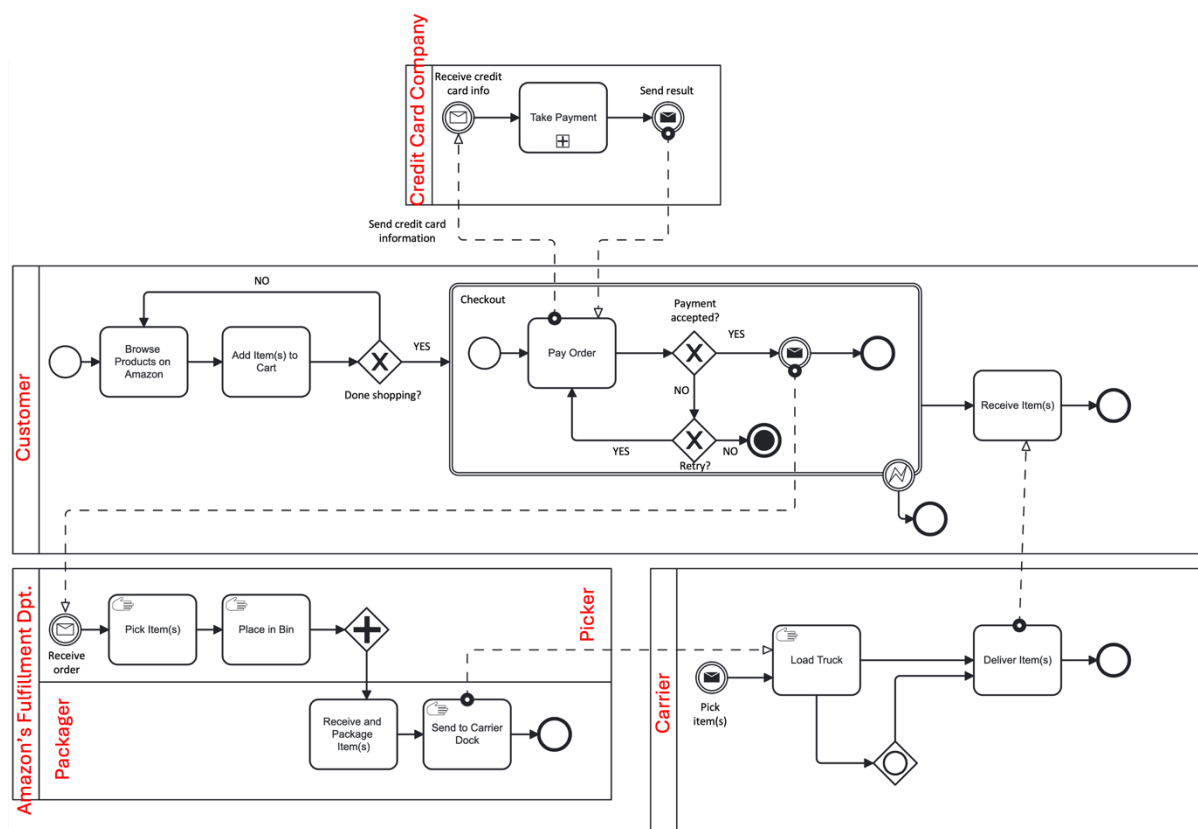
Task 1 (25 points)

1.1 (5 points)

In a BPMN model, pools represent the major participants (often distinct organizations or systems) in the process, while lanes represent roles, departments, or specific functions within an organization or system. Together, pools and lanes help delineate distinct roles and responsibilities within the overall process.

In the BPMN model of the online shopping and delivery process, the names of the pools and lanes are missing. Assign appropriate names to them (based on the description of the case) to ensure clarity in the workflow and help identify points of interaction between participants. Indicate where in the model are those pools and lanes located.

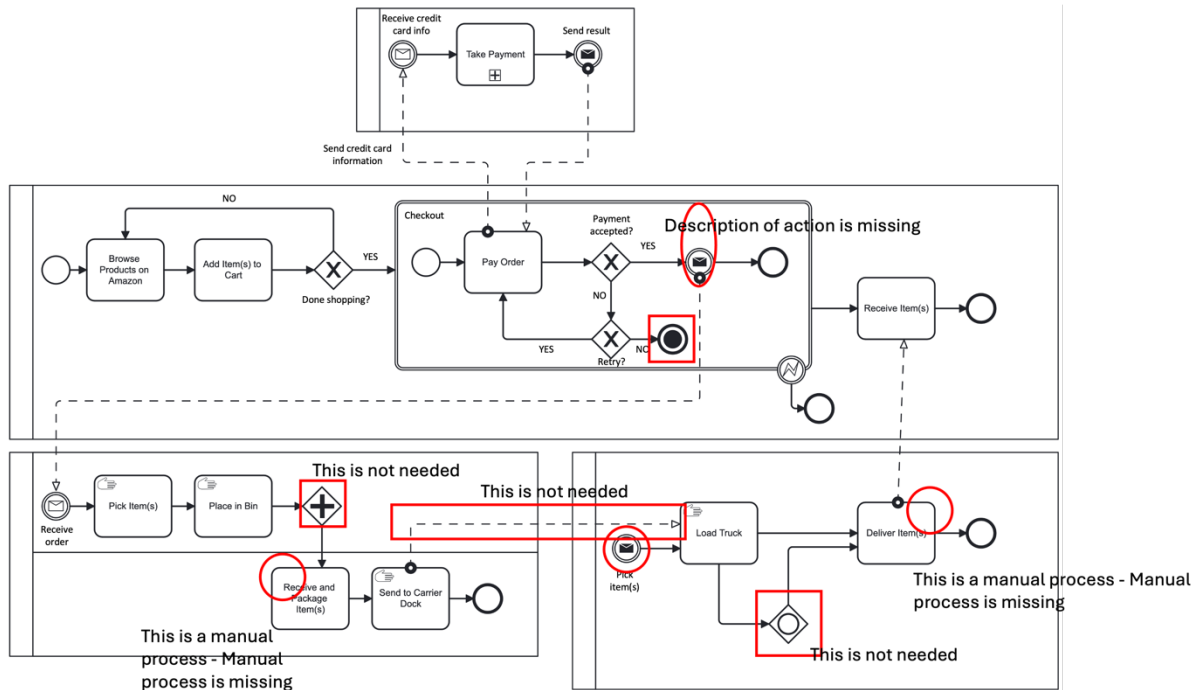
Keep your answer between 0-150 words.



1.2 (20 points)

The BPMN model contains several mistakes, both syntactic and semantic. Find at least two syntactic and two semantic mistakes (so, four mistakes in total). Describe where in the model those mistakes are located. Provide an explanation for why you consider each to be a mistake. **Note:** the missing names of pools and lanes from 1.1. do not count as mistakes.

Keep your answer between 0-250 words.



- there is no end event for the "Credit Card Company" pool, which should signify the completion of the payment process after sending the result.
- Wrong gateways and wrong use of gateways.
- missing notations.
- carrier task begins with send message, while it should begin with receive message
- the arrow from the "send to carrier dock" in the amazon's pool should go to the "pick items" message in the carrier's pool

Task 2 (20 points)

As explained, the human roles play a critical part in ensuring the smooth processing and fulfillment of the customer's order.

2.1 (15 points)

What types of Business Information Systems do you think are needed to support the task of each of the following participants?

- Credit card company;
- Amazon fulfillment department;
- Carrier dock

Name two types of IS per type of participant, so **for each participant you get 2 points**. Elaborate on your answers. **For each participant you get 3 points**.

Keep your answer between 0-250 words.

- Credit card company:** (i) Transaction Processing System (TPS) – explanation: Manages and executes credit card transactions in real-time, ensuring the payment information provided by the customer is processed, authorized, or declined accurately; (ii) Customer Relationship Management System (CRM) – explanation: Manages customer accounts, including transaction history and service issues, ensuring a smooth customer experience.

- (b) **Amazon fulfilment department:** (i) Order Processing System - explanation: Coordinates incoming orders, directing pickers and packagers on what items to prepare for each order. It helps manage order details and ensures that each customer's order is correctly handled through the fulfillment process; (ii) Warehouse Management System (WMS) - Part of a Supply Chain Management System (SCM) – explanation: Tracks inventory, optimizes item locations within the warehouse, manages picking routes, and oversees the storage and retrieval processes. It guides the picker to the correct items and helps streamline warehouse efficiency.
- (c) **Carrier dock:** (i) Logistics and Distribution Planning - Part of a Supply Chain Management System (SCM) – explanation: Manages the coordination of shipments, schedules pickups, and tracks the movement of packages within the carrier's network. This system helps ensure that packages are picked up on time and assigned to the correct delivery routes; (ii) Tracking System - Part of a Supply Chain Management System (SCM) – explanation: Provides real-time updates on the status of packages, tracking their journey from the carrier dock to the customer. It allows both Amazon and customers to monitor the location and estimated delivery time of their orders.

2.2 (5 points)

A Fraud Detection System (FDS), which is a type of DSS, uses data analysis to detect and flag unusual transaction patterns that might indicate fraudulent activities, helping to secure transactions and minimize risk. Which of the above participants needs an FDS – in addition to the two other types you identified in 2.1 (2 points)? Explain your answer (3 points). **Note:** you cannot use the FDS as answer to 2.1.

Keep your answer between 0-150 words.

The **Credit Card Company** would need a Fraud Detection System (FDS) in addition to the two other systems identified earlier (TPS and CRM).

Explanation: Credit card companies process thousands of transactions daily, making them vulnerable to fraudulent activities. An FDS can analyze transaction patterns, such as unusual spending amounts, geographic inconsistencies, or deviations from a customer's typical behavior, to identify potentially fraudulent activities in real time. By flagging suspicious transactions, the system helps secure payments and protects both the customer and the credit card company from financial losses, thereby enhancing trust and operational security.

Task 3 (10 points)

3.1 (5 points)

The differentiation between data, information, and knowledge helps each participant make better decisions at every stage of the order, fulfillment, and delivery process in the case described above. What could be data, information, and knowledge in the context of the Amazon shopping and delivery process? Give two examples of data, two examples of information and one example of knowledge for any of the tasks described in the case. **Each of your answers gets 1 point.**

Keep your answer between 0-150 words.

Data are raw facts and figures that are unprocessed and have not been given any specific meaning. In this scenario, data could include:

- Individual product IDs and prices.
- Customer ID numbers.

- Timestamp of when items were added to the cart.
- Payment details like card number, expiration date, and transaction amount.
- Warehouse inventory counts.

Information is processed or organized data that provides meaning. It gives context to raw data, making it more useful for specific purposes. In this example, information could include:

- A summary of the customer's order, including items, quantities, and total price.
- Confirmation of payment authorization or rejection from the credit card company.
- A list of items to be picked and packed, including their locations in the warehouse.
- Delivery tracking updates, showing the package's current location and estimated arrival time.

Knowledge is derived from information through understanding, insights, or experience, allowing informed decisions. It involves the application of information in a way that provides value. In this scenario, knowledge could include:

- Knowing which products are frequently ordered together, enabling Amazon to make recommendations.
- Understanding that certain patterns in payment rejections may indicate fraud, which the Fraud Detection System uses to flag risky transactions.
- Insights into peak order times, allowing the fulfillment department to optimize staffing and resource allocation.
- Recognizing that certain routes are prone to delivery delays, enabling the carrier to adjust logistics accordingly.

3.2 (5 points)

We talked about the dimensions of valuable information. List 3 that you believe are relevant for the case of credit card transactions in the above case (**1 point for each**) and briefly explain and justify why two of them are important (**1 point for each**).

Keep your answer between 0-150 words.

Accurate: Accuracy is crucial in credit card transactions because any errors (e.g., incorrect payment amount or misinterpretation of transaction data) could lead to disputes, financial losses, or customer dissatisfaction. Accuracy ensures that transactions are processed correctly, reflecting the exact amount and details of each purchase.

Secure: Security is essential for credit card transactions, as these involve sensitive financial information. Protecting customer data from unauthorized access helps prevent fraud, maintains customer trust, and ensures compliance with regulations.

Any other dimension is acceptable if justified sufficiently.

Task 4 (5 points)

Amazon's online shopping and delivery process involves multiple participants, each relying on accurate, real-time data to perform their tasks. Currently, these participants operate with limited visibility into the overall workflow, leading to possible inefficiencies such as delays in order fulfillment, mismatched inventory levels, and inconsistent delivery updates. Which kind of information system (**except from an ERP**) should be used to for real-time monitoring and reporting across its shopping and delivery processes, using available transaction information? Motivate your answer.

Keep your answer between 0-50 words.

Management Information System (MIS) – we emphasized that MIS use TPS information to provide insight and reports. ERP is an alternative answer, although the keywords here are “reporting” and “using TPS”.

Task 5 (15 points)

Amazon emphasizes efficiency, scalability, and customer satisfaction in its online shopping and delivery processes. Therefore, the organization is considering automating warehouses with robots, or experimenting with drone deliveries.

5.1 (10 points)

Those are major organizational changes. We have discussed two ways of doing this: re-engineering and continuous improvement. Which of these two approaches would you consider to be most suitable for the above changes? Elaborate on your answer, also explaining why the other option is less suitable. *Keep your answer between 0-150 words.*

Re-engineering would be the most suitable approach. This is because these changes represent major shifts in Amazon’s operational processes, requiring a fundamental redesign. Implementing robots and drones would require Amazon to redesign its logistics, inventory management, and delivery processes to accommodate these new technologies.

While continuous improvement is valuable for refining existing processes, it is less effective for achieving the kind of dramatic transformations described. For example, incremental changes to warehouse operations (e.g., optimizing current manual picking processes) would not result in the efficiency gains provided by robotic automation. Also, small adjustments to delivery logistics are far from the revolutionary potential of drone deliveries in reducing costs and delivery times.

5.2 (5 points)

Those are not only major organizational changes, but also innovations. We discussed two types of innovations. Which type of innovation is the warehouse automation and which type of innovation is the drone delivery? Elaborate on your answers.

Keep your answer between 0-150 words.

Warehouse automation with robots can be considered incremental innovation because it enhances the current warehouse processes, such as picking, sorting, and packing, which are already established. Robots are integrated into these processes to improve efficiency without fundamentally altering the workflow. It focuses on improving productivity, reducing manual labor, and minimizing errors, representing a continuous improvement of existing operations rather than a groundbreaking change.

Drone delivery, on the other hand, is a radical innovation because it introduces a transformative way of delivering packages, fundamentally altering the traditional delivery model (trucks, vans, etc.). Drone delivery requires new infrastructure, regulatory adjustments, and advanced technology, making it a disruptive innovation that could redefine last-mile delivery.

Task 6 (5 points)

Amazon’s online shopping and delivery is a Business-2-Consumer (B2C) e-commerce activity. We discussed three additional different types of e-commerce. List them and for each, give a real-life example.

Keep your answer between 0-50 words.

Types of Ecommerce

FedEx



Business to Business

NETFLIX



Business to Consumer

airbnb

ebay



Consumer to Consumer



Consumer to Business



freelancer